CODI-GPT-2 · Context isolation (6 methods)

Question

If we subtract the 'context' part of the activation, does the math signal (operator + digits) survive? Different definitions of 'context' give different answers. We try 6.

Setup

Activations: 1000 SVAMP × 16 decode positions × 13 layers × 768.
For each method we apply a residualization, then refit per-(pos, layer) probes for {units, tens, operator} (5-fold CV) using model-emitted labels.

Punchline

- Method A/C (subtract per-op mean) — removes operator entirely (op→53% at template positions, partial elsewhere) but leaves digit info intact.
- Method B (top-80%-var orthog) — too aggressive, also removes digits at later positions because the high-variance directions ARE the math.
- Method D (grand mean) — no-op (probe is invariant to mean shift).
- Method E (top-3-PC orthog) — clean 'remove context, keep math'. All three signals survive within 1-6 pp of baseline at every position.

Six context-isolation methods

Goal: subtract 'context' from each (pos, layer) activation, then re-fit operator/units/tens probes. Compare to baseline to see what each method actually removes.

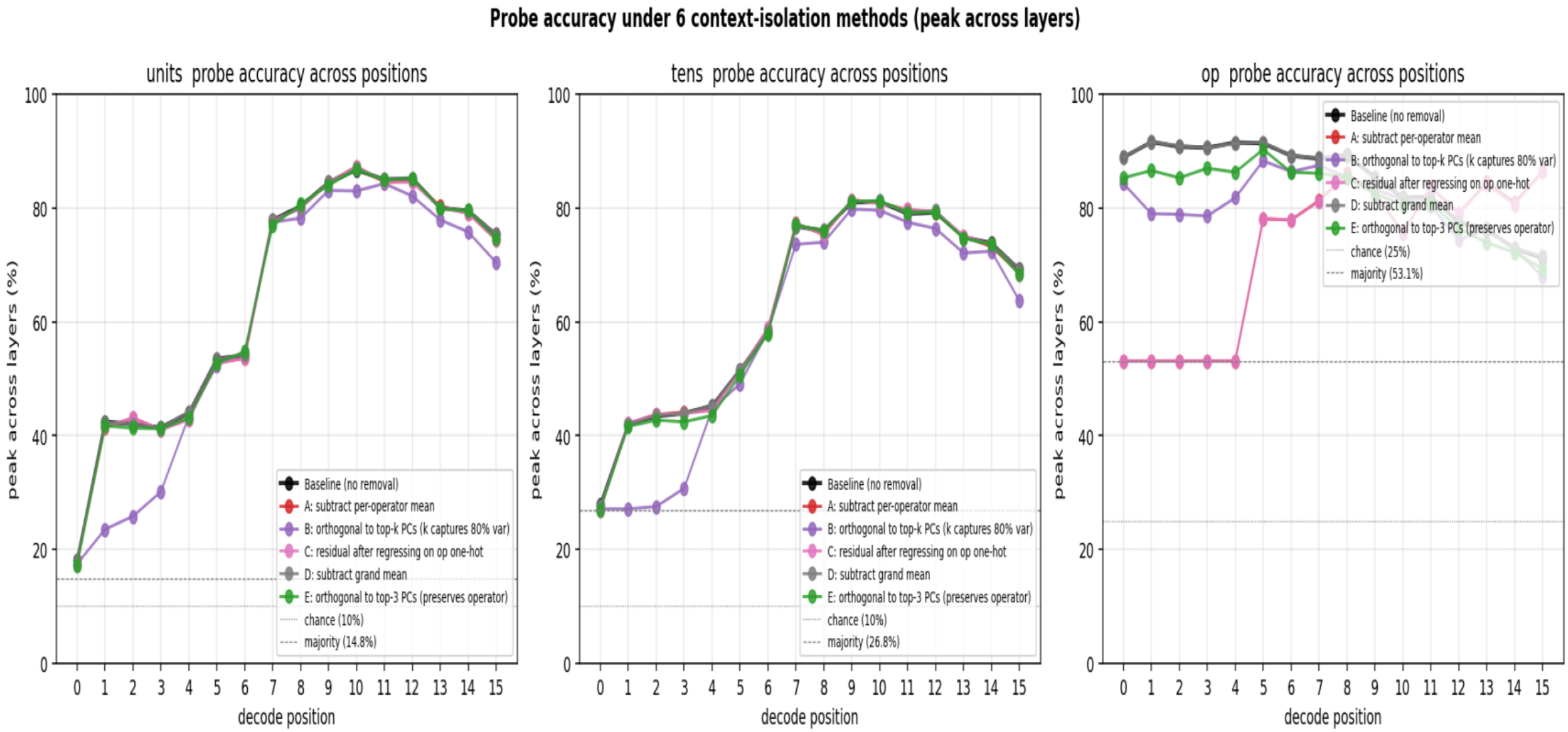
Method definitions

- baseline: no removal (raw activations).
- A: subtract per-operator-class mean μ_c[p, l]. Removes the additive operator signal explicitly.
- B: PCA orthogonal complement keeping only the top-k PCs that capture 80% of variance. k = 4 at pos 0; up to 20 at pos 1; only 1-2 at pos 4-15. Aggressive removal.
- C: residual after Ridge regressing each activation dim on a 4-D operator one-hot. Mathematically equivalent to A.
- D: subtract grand mean (zero-center). No information removal — probes use StandardScaler internally so this is a no-op for probe accuracy. Sanity check.
- E: orthogonal to TOP-3 PCs only. Removes the biggest shared modes (likely template/context) while leaving smaller-variance operator/digit directions intact.

What 'removing context but preserving operator+digit' looks like

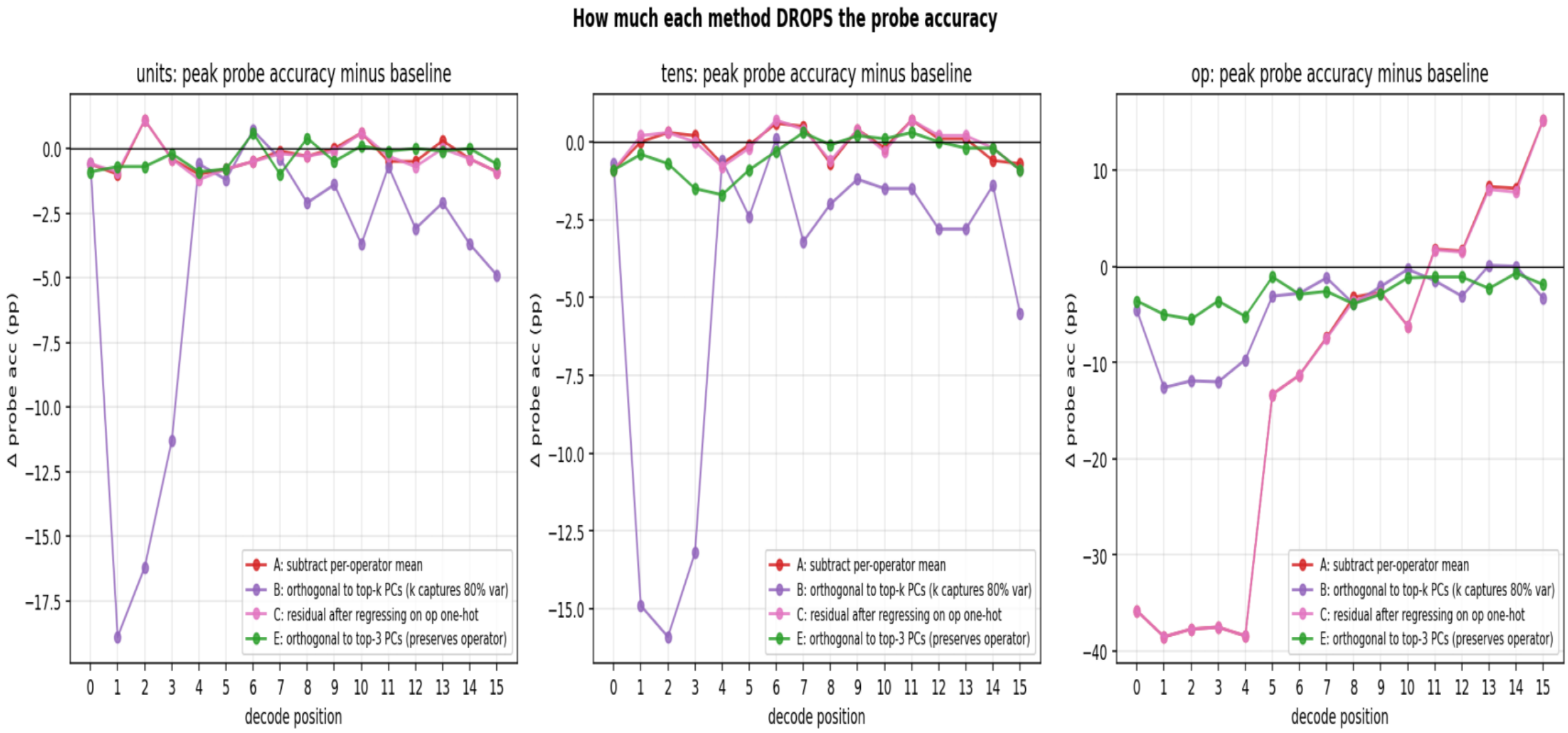
Method E is the clean version. At every position, op stays >80% (pos 0-9), units/tens drop by <2 pp, but the top 3 'shared' PCs are gone.
If the model relied on those PCs for prediction, probes would have collapsed; they don't.

Probe accuracy under each method (raw)



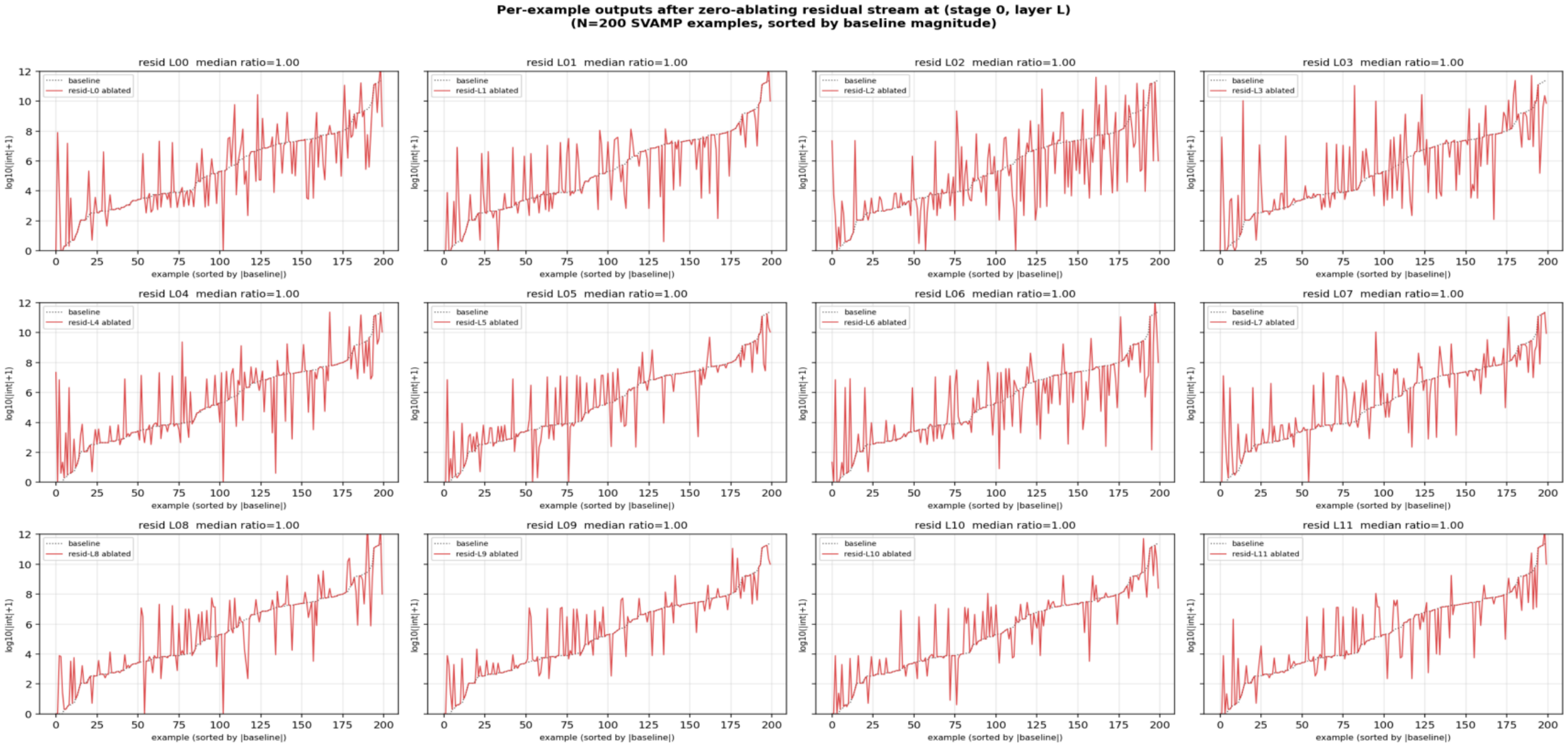
E (green) tracks baseline closely (removes only the top-3 highest-variance modes). Method A/C (red/pink) zaps op at template positions (drops to majority baseline). Method B (purple) over-prunes everything once

Drop relative to baseline (pp)



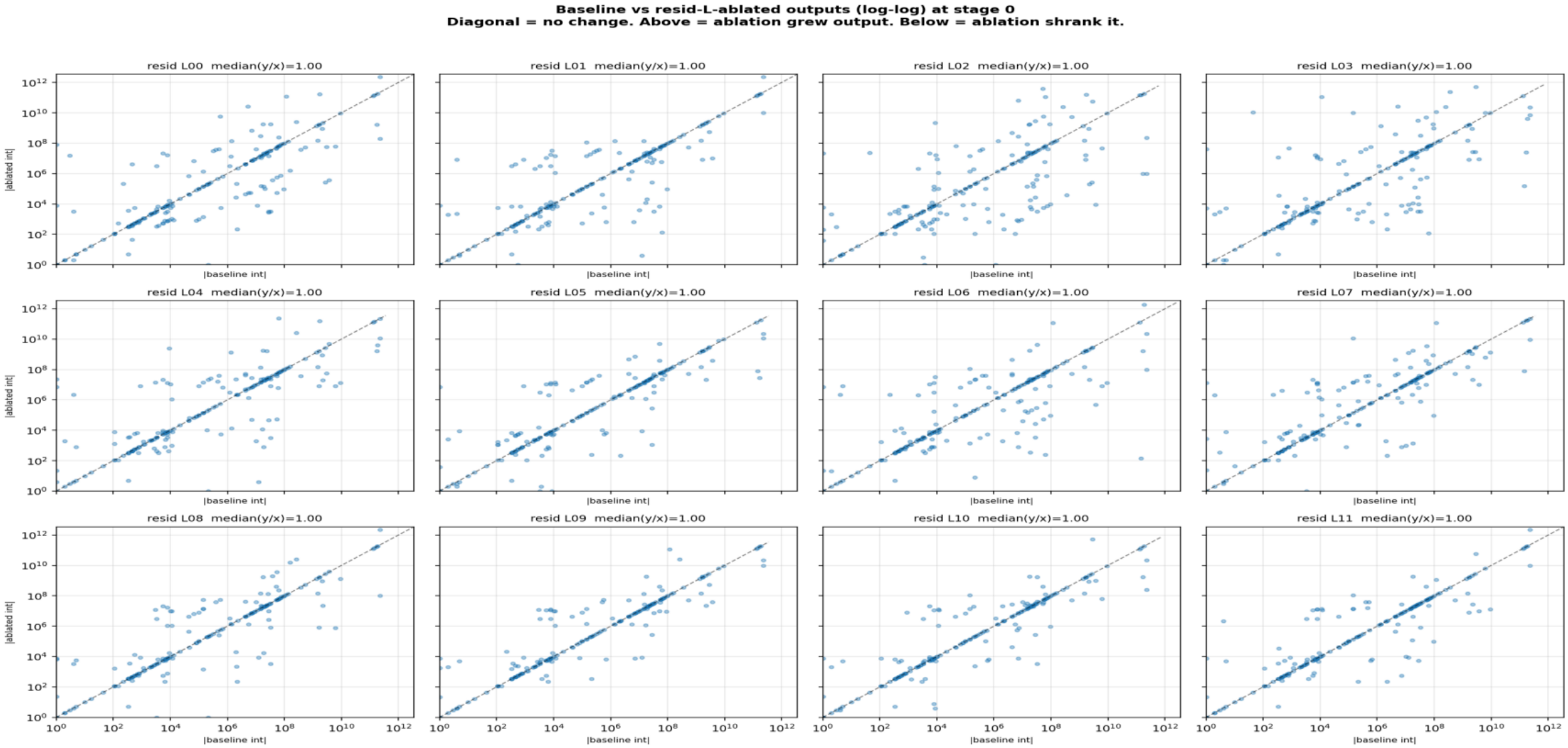
Method E (green) stays within ±5 pp of baseline at every position. Method A (red) hits operator hard at pos 0-4. Method B (purple) damages units/tens at pos 1-2 because top-PCs there encode the math.

Per-example ablation: residual at (stage 0, layer L)



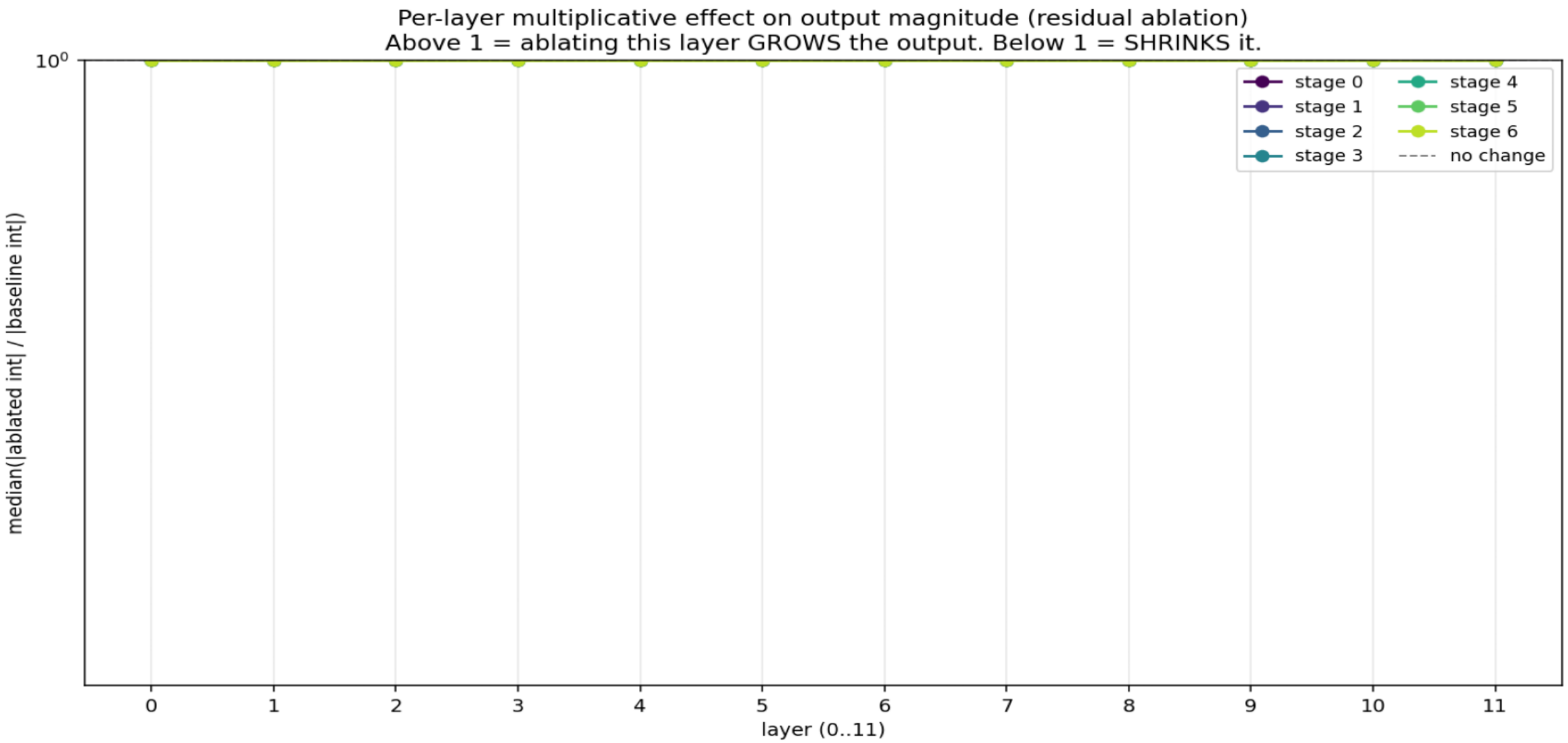
agnitude. Black dotted = baseline |output|. Red = output after zero-ablating residual at that layer (stage 0 = prompt position). L2 systematically pulls the red trace below baseline (shrinks output); L7 pushes it ab

Per-example ablation: scatter plot



x = |baseline|, y = |after-resid-L-ablation|, log-log. Diagonal = no change. L2: most points BELOW diagonal (ablation shrinks). L7: most ABOVE (ablation grows). Per-layer median ratios in each title.

Per-layer multiplicative effect summary



Median(|ablated| / |baseline|) per layer, colored by stage. Layer 2 consistently around 0.3-0.6× across all 7 stages. Layer 7 consistently 5-7× across all stages. These are stable per-layer effects, not noise.

Synthesis

What context-isolation tells us

- The model's residual at template positions (pos 0-3) carries operator info that is mostly 'additive' across operator classes (Method A removes it cleanly; Method E barely touches it).
- At answer positions (pos 5+), op probe is more entangled with digit info — Method A only partially removes it (drops to 78-86%).
- The TOP-3 highest-variance PCs at each (pos, layer) are mostly 'context'. Removing them barely hurts any probe → operator and digit info live in smaller-variance directions, not in the dominant modes.

What per-example ablation tells us

- Residual ablation at different layers has DIRECTIONAL effects on output magnitude. L2 systematically shrinks; L5/L7/L8 systematically grow.
- This is a stable per-layer property: the same direction across all 200 examples and all 7 stages.
- Implication: the model's layers contribute structured 'magnitude tweaks' that compose into the final output. Removing one tweak biases the result one way; removing another biases it the opposite way.

Open question

Which layer's contribution carries the operator vs the magnitude vs the digit? The current ablation collapses to net magnitude shift; a finer experiment would track which probe (op/units/tens) is restored after each single-layer ablation.